

Jahanvi Patel

Data Scientist

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PROFESSIONAL SUMMARY

Data Scientist with **4+ years** of experience in designing scalable machine learning systems, real-time data pipelines, and **NLP solutions** across finance and technology sectors. Proven expertise in **Python, Spark, TensorFlow, Scikit-learn**, and **cloud platforms** such as **AWS (S3, Glue, EMR, Kinesis)**. Adept at building and optimizing **ML pipelines** using **MLOps tools (MLflow, Kubernetes)** and delivering actionable insights through advanced dashboards (**Tableau, Cognos**). Successfully applied **LSTMs, Transformers**, and **Generative AI** to enhance asset **prediction models** and **automate document** processing. Strong track record of improving **data efficiency, forecasting accuracy**, and **operational decision-making** in dynamic enterprise environments.

PROFESSIONAL EXPERIENCE

JPMorgan | NY | Data Scientist

October 2023 – Present

- Designed and deployed **scalable, real-time data pipelines** using **AWS services (S3, Glue, EMR, and Kinesis)** to process market risk data, improving data processing efficiency by **50%** and enabling instant analytics for risk monitoring dashboards.
- Applied cutting-edge **deep learning architectures**, including **LSTMs** and **Transformer networks**, within the firm's proprietary asset price **prediction engine**, achieving a **35%** improvement in forecasting accuracy.
- Optimized data processing pipelines using **Spark** and **Spark Streaming**, applying **Spark MLlib** for machine learning model training, which improved real-time **data analysis** performance by **40%** in dynamic environments.
- Pioneered the development and **deployment of Natural Language Processing (NLP) models** using **Python, NLTK**, and **spaCy** to extract actionable insights from unstructured regulatory filings and financial news.
- Orchestrated containerized data workflows with **Kubernetes** and automated **ETL pipelines**, integrating **CI/CD processes**, monitoring tools, security protocols, performance tuning, and logging mechanisms, ensuring seamless data integration and reducing deployment times.
- Created **interactive real-time dashboards** in **Tableau**, transforming **complex datasets** from **credit risk models** into **accessible visual insights**, which empowered **risk, compliance, and trading teams** to make **timelier, data-driven decisions**.
- Integrated **Generative AI techniques** for synthetic data generation and automated document processing, enhancing **predictive analytics pipelines** and reducing manual workload in financial document review by over **40%**.

IBM | India | Data Scientist

November 2019 – July 2022

- Developed and deployed **machine learning** models using **Python, SciPy, Scikit-learn**, and **TensorFlow**, achieving a **20%** improvement in predictive accuracy for business-critical applications.
- Created interactive dashboards with **IBM Cognos Analytics**, integrating **predictive model outputs** to improve **sales forecasting** in the retail analytics division, contributing to a **15%** increase in data-driven decisions by business stakeholders.
- Designed and executed **A/B testing frameworks** to evaluate the performance of **machine learning models** in production, enabling data-driven decisions on model deployments and feature rollouts.
- Optimized data pipelines by redesigning **SQL queries** and **schemas** in **IBM Db2**, reducing data retrieval time by **25%** in a high-volume financial transaction monitoring system, enabling faster **anomaly detection**.
- Built end-to-end **MLOps pipelines** using **MLflow** and **Kubernetes** to automate model training, deployment, and monitoring, ensuring scalable and reproducible machine learning workflows in production.
- Leveraged **Python** and **deep learning** techniques for **Natural Language Processing (NLP)** tasks, achieving a **35%** improvement in sentiment analysis accuracy and automating text data extraction processes.

TECHNICAL SKILLS

Programming Languages: Python (NumPy, Pandas, Matplotlib, SciPy, Scikit-learn, TensorFlow, PyTorch), R (ggplot2, dplyr, caret), SQL (MySQL, PostgreSQL, SQL Server), Scala/Java (for big data technologies)

Data Manipulation & Analysis: Data Wrangling & Cleaning, Feature Engineering, Statistical Analysis, Exploratory Data Analysis (EDA)

Machine Learning & AI: Supervised & Unsupervised Learning, Deep Learning (Neural Networks, CNNs, RNNs), Natural Language Processing (NLP), Time Series Analysis, Model Evaluation & Tuning (Cross-validation, Hyperparameter Tuning)

Data Visualization & Big Data Technologies: Tableau, Power BI, Seaborn, Plotly, Matplotlib, Hadoop, Spark, Hive, HBase

Cloud Platforms & Data Warehousing: AWS (S3, Redshift, SageMaker, Lambda), Azure (Azure Data Lake, Azure Machine Learning)

Databases: SQL (MySQL, PostgreSQL), NoSQL (MongoDB, Cassandra), Data Warehousing (Snowflake, Redshift, BigQuery)

Tools & Frameworks: Jupyter Notebooks, Apache Airflow (for scheduling), Git/GitHub (version control), Docker/Kubernetes (for deployment)

Business & Domain Knowledge: A/B Testing & Experimentation, Business Intelligence (BI) tools, Data-Driven Decision Making, Industry-specific knowledge (Finance, Healthcare, E-commerce, etc.)

Soft Skills: Strong Communication & Presentation Skills, Critical Thinking & Problem Solving, Stakeholder Collaboration

EDUCATION

Master's in Information Technology and Management | The University of Texas at Dallas, TX, USA.

Bachelor's in Computer Engineering | Ganpat University, Mehsana, Gujarat, India.